

Substitutions D.E :

①

② Homogeneous D.E :

This D.E's is of the form:

$$y' = F\left(\frac{y}{x}\right) \text{ --- (1)}$$

Methodology: our goal is to Convert equ(1) into separable D.E.

step 1) let $v(x) = \frac{y}{x}$; then

$$y = x \cdot v \text{ --- (2)}$$

step 2) Differentiate equ (2) to get:

$$(3) \text{ --- } y' = v + x v' \text{ (Product rule)}$$

step 3) substitute equ (*) and equ (3) into equ (1) to get :

$$v + x v' = F(v) \cdot \text{ But } v' = \frac{dv}{dx}$$

$$\text{then } v + x \frac{dv}{dx} = F(v) \Rightarrow x \frac{dv}{dx} = F(v) - v$$

$$\text{Thus, } x dv = (F(v) - v) \cdot dx \Rightarrow dv = \frac{(F(v) - v)}{x} dx \Rightarrow$$

$$\boxed{\frac{dv}{F(v) - v} = \frac{dx}{x}} \text{ separable D.E .}$$

EX: #1) Solve the following IVP:

(2)

$$xy y' + 4x^2 + y^2 = 0, \quad y(2) = -7, \quad x > 0$$

note that: Each term in the D.E. have the same order.

Solution First, Divide both sides by x^2 to get:

$$\frac{y}{x} y' + 4 + \frac{y^2}{x^2} = 0 \Rightarrow$$

$$\frac{y}{x} y' = -4 - \left(\frac{y}{x}\right)^2, \quad \text{Let } \boxed{v = \frac{y}{x}}, \text{ then.}$$

$$v(v + xv') = -4 - v^2, \text{ then}$$

$$\text{Note } \boxed{y' = v + xv'}$$

$$v^2 + xv v' = -4 - v^2.$$

Rewrite the D.E. to get every-thing separated out:

$$vxv' = -4 - 2v^2, \quad \text{Divide by } \underline{v} \text{ to get:}$$

$$xv' = \frac{-4 - 2v^2}{v}$$

$$x \frac{dv}{dx} = -\frac{4 + 2v^2}{v}, \quad \text{multiply by } dx \text{ and divide by } \underline{x}$$

$$\Rightarrow dv = -\frac{1}{x} \left(\frac{4 + 2v^2}{v} \right) dx. \quad \text{multiply by } \frac{v}{4 + 2v^2} \Rightarrow$$

$$\Rightarrow \frac{v}{4 + 2v^2} dv = -\frac{1}{x} dx \quad (\text{separable D.E.})$$

Integrate both sides

$$\frac{1}{4} \int \frac{4v}{4 + 2v^2} dv = \int -\frac{1}{x} dx$$

(3)

$$\frac{1}{4} \int \frac{du}{u} = -\ln(x) + C$$

$$\frac{1}{4} \ln(4+2v^2) = \ln(x)^{-1} + C \Rightarrow \ln(4+2v^2)^{\frac{1}{4}} = \ln(x)^{-1} + C$$

Take e to both sides to get:

$$(4+2v^2)^{\frac{1}{4}} = e^C \cdot e^{\ln(\frac{1}{x})} \Rightarrow (4+2v^2)^{\frac{1}{4}} = \frac{e}{x} \Rightarrow (4+2v^2) = \left(\frac{e}{x}\right)^4$$

But $\boxed{v = \frac{y}{x}} \Rightarrow$

$$\boxed{v^2 = \frac{1}{2} \left[\frac{e}{x^4} - 4 \right]}$$

$$\frac{y^2}{x^2} = \frac{1}{2} \left(\frac{C-4x^4}{x^4} \right), \text{ multiply by } x^2 \Rightarrow$$

$$y^2 = \frac{1}{2} x^2 \left(\frac{C-4x^4}{x^4} \right) = \frac{C-4x^4}{2x^2} \Rightarrow$$

$$y^2 = \frac{C-4x^4}{2x^2}$$

But $y(2) = -7$

$$49 = \frac{C-4(16)}{2(4)} \Rightarrow \boxed{C = 456}$$

$$\text{So, } y^2 = \frac{456-4x^4}{2x^2} = \frac{\cancel{2}(228-2x^4)}{\cancel{2}x^2} = \frac{228-2x^4}{x^2}$$

$$\Rightarrow y(x) = \pm \sqrt{\frac{228-2x^4}{x^2}}, \text{ since } y(2) = -7, \text{ then we choose}$$

$$\therefore \boxed{y(x) = -\sqrt{\frac{228-2x^4}{x^2}}}$$

X#2) Solve the IVP

(4)

$$xy' = y(\ln(x) - \ln(y)) \quad , \quad y(1) = 4, \quad x > 0$$

Solution Note each term in the D.E have the same order.

Now, our goal is to Convert this Homogeneous D.E into separable D.E:

$$y' = \frac{y}{x} \left[\ln\left(\frac{x}{y}\right) \right] \quad , \quad \text{Let } \boxed{v = \frac{y}{x}} \text{ and } \boxed{\frac{1}{v} = \frac{x}{y}}$$

$$v + xv' = v \ln\left(\frac{1}{v}\right) \Rightarrow xv' = v \ln\left(\frac{1}{v}\right) - v \Rightarrow$$

$$\Rightarrow x dv = v \left[\ln\left(\frac{1}{v}\right) - 1 \right] dx \quad , \quad \text{divide by } x \text{ both sides and divide by } v(\ln(\frac{1}{v}) - 1)$$

$$\frac{dv}{v(\ln(\frac{1}{v}) - 1)} = \frac{dx}{x} \quad , \quad \text{but } \ln\left(\frac{1}{v}\right) = -\ln(v), \text{ since } \ln(1) = 0$$

$$\frac{dv}{-v(\ln(v) + 1)} = \frac{dx}{x} \Rightarrow \frac{dv}{v(\ln(v) + 1)} = -\frac{dx}{x}$$

Integrate both sides to get:

$$\int \frac{dv}{v(\ln(v) + 1)} = \int -\frac{1}{x} dx$$

$$\text{Let } u = \ln(v) + 1.$$

$$du = \frac{1}{v} dv.$$

$$\text{So, } \int \frac{du}{u} = -\ln(x) + C \Rightarrow \ln|u| = -\ln(x) + C.$$

$$\Rightarrow \ln(\ln(v) + 1) = -\ln(x) + C.$$

Take e to both sides:

$$\ln(v) + 1 = e^{\frac{C}{x} - \ln(x)}$$

$$\ln(v) + 1 = \frac{C}{x} \Rightarrow \ln(v) = \frac{C}{x} - 1.$$

$$\Rightarrow v(x) = e^{\frac{C}{x} - 1}$$

$$\frac{y}{x} = e^{\frac{C}{x} - 1}$$

$$y(x) = x e^{\frac{C}{x} - 1}, \quad \text{But } y(1) = 4, \text{ then}$$

$$4 = y(1) = 1 \cdot e^{C-1} \Rightarrow \ln(4) = C-1 \Rightarrow C = \ln(4) + 1$$

$$\therefore y(x) = x e^{\frac{\ln(4) + 1}{x} - 1}$$